Testing a Global CRC Microbial Consortium:

Independent Re-analysis of the MAL2 Malaysian Cohort

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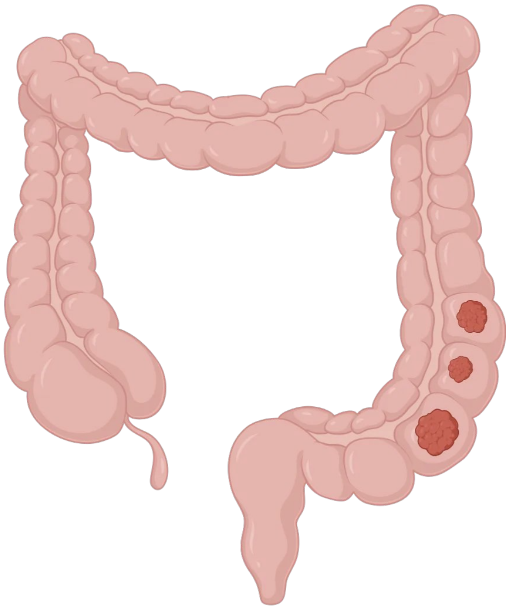
BIOMI 4300

01
Introduction

"CRC Is the 3rd Most Common Cancer Globally, Yet Non-Genetic Risk Factors Remain Poorly Characterized"

- ~1.93 million new CRC cases globally in 2022; 9.6% of all cancers; 2nd leading cause of cancer death
- Only ~5–10% of CRC is attributable to hereditary syndromes (APC mutations → Familial Adenomatous Polyposis; Lynch syndrome → MLH1/MSH2)
- The remaining ~90% of sporadic cases implicate environmental/microbial drivers, including the **gut microbiome** as a potential contributing factor



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"Multiple Studies Confirm That CRC Tissue Is Consistently Enriched for Oral Pathogens and Depleted of Protective Commensals"

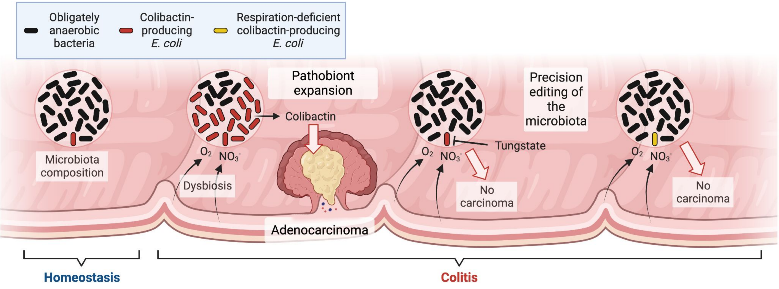
Over a dozen studies using stool and tumor tissue sequencing have associated CRC with microbial dysbiosis in taxonomic composition and/or mucosal structural organization

Consistently enriched in CRC tissue:

- Fusobacterium nucleatum (oral pathogen; promotes tumorigenesis via FadA/Fap2 adhesins)
- Bacteroides fragilis (ETBF strain) (encodes the BFT metalloprotease toxin)
- Parvimonas micra, Peptostreptococcus stomatis, Gemella morbillorum (oral anaerobes)

Consistently depleted in CRC tissue:

- Faecalibacterium prausnitzii (anti-inflammatory butyrate producer)
- Bacteroides vulgatus, B. dorei, B. stercoris (protective commensals)



https://www.pnas.org/doi/10.1073/pnas.2316579120

npj | Biofilms and Microbiomes

www.nature.com/npjbiofilms

Corrected: Author Correction

ARTICLE OPEN

High-resolution bacterial 16S rRNA gene profile meta-analysis and biofilm status reveal common colorectal cancer consortia

Julia L. Drewes[1], James R. White[2], Christine M. Dejea[1], Payam Fathi[1], Thevambiga Iyadorai[3], Jamuna Vadivelu[3], April C. Roslani[3], Elizabeth C. Wick[1], Emmanuel F. Mongodin[4], Mun Fai Loke[3], Kumar Thulasi[3], Han Ming Gan[5], Khean Lee Goh[3], Hoong Yin Chong[3], Sandip Kumar[3], Jane W. Wanyiri[1] and Cynthia L. Sears[1]

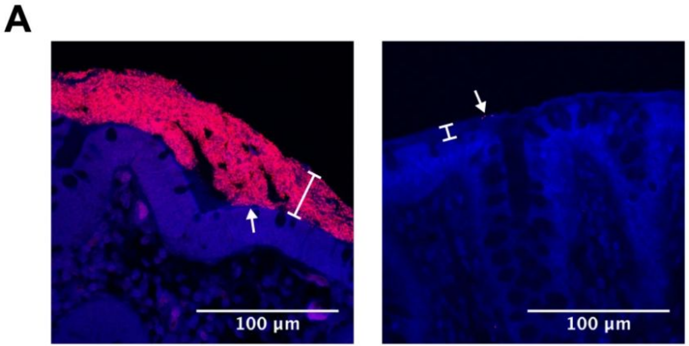
"Polymicrobial Biofilms Cover ~100% of Right-Sided CRC Tumors and Their Paired Normal Tissue Suggesting They Precede Tumorigenesis"

Biofilms = dense polymicrobial aggregates invading the inner mucus layer; visualized by FISH (Eub338 probe)

Right-sided tumors: 100% biofilm-positive across all three cohorts (USA, MAL1, MAL2)

Left-sided tumors: Only 37.5% biofilm-positive in MAL2 (Fisher's exact p = 0.008)

Biofilm status of paired normal tissue matches the tumor in 20/23 MAL2 patients → **biofilm precedes cancer**



a Microbial biofilms in Carnoy's-fixed tissue were detected by FISH with the universal 16S rRNA gene probe Eub338 (stained in red) and the nucleic acid stain DAPI (blue).

"Genus-Level Resolution and Cross-Lab Heterogeneity Masked Species-Specific CRC Signals Across 11+ Prior Datasets"

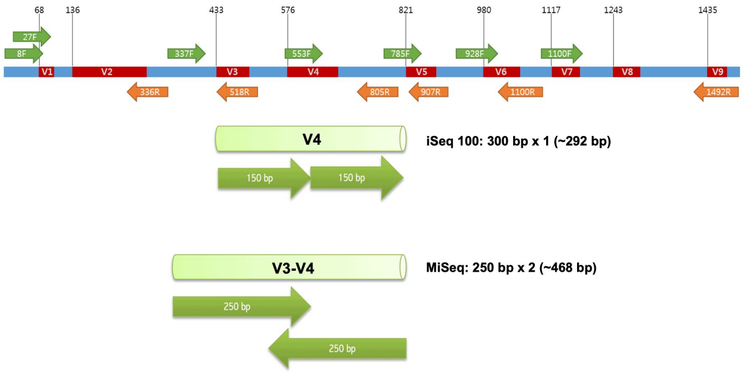
Different primers (V1-V2, V3-V4, V3-V5), platforms (454, MiSeq), and extraction methods → contradictory findings

OTU-level clustering conflates pathogenic and commensal species within the same genus

Canonical example: F. nucleatum (CRC-enriched) vs. F. mortiferum/varium (not enriched) are invisible at genus level — their signals cancel

F. prausnitzii* (commensal, CRC-depleted) was historically misclassified as Fusobacterium — reversing the signal entirely

Solution: Drewes et al. applied Resphera Insight (species-level 16S classifier) uniformly to all 12 datasets*



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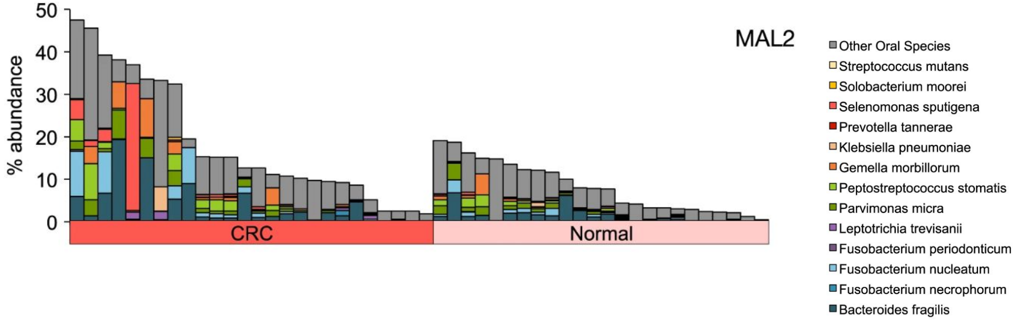
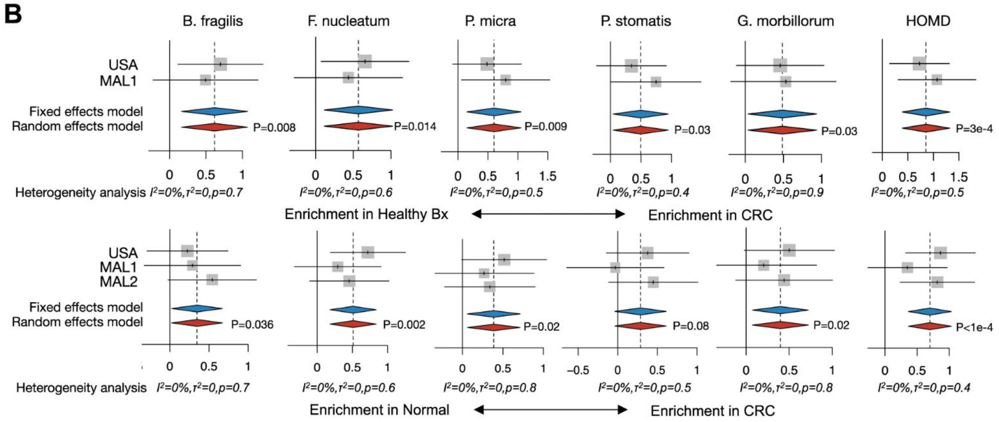
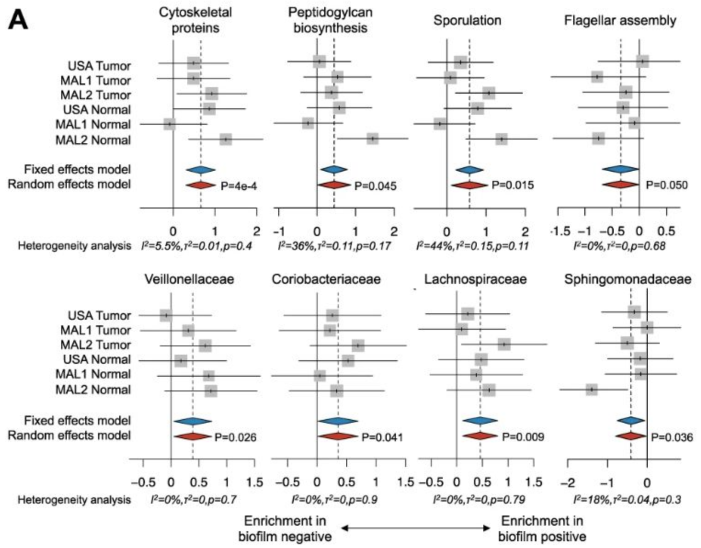
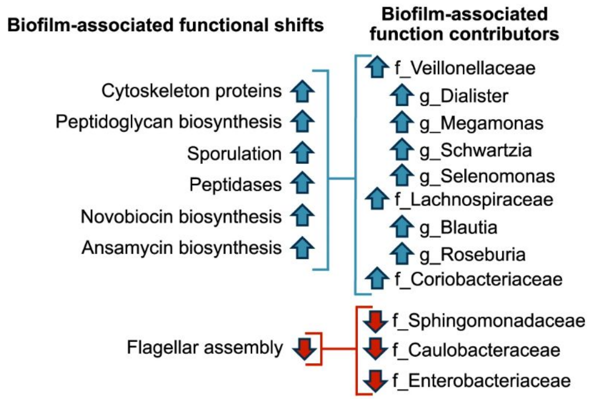
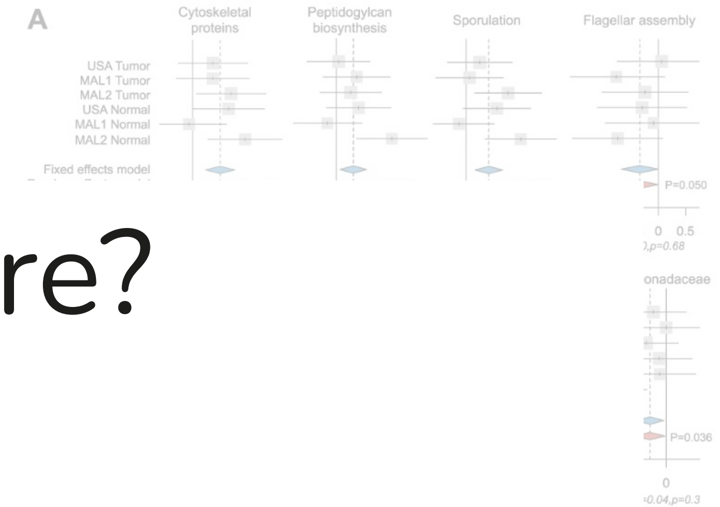




Associations of *B. fragilis* and human oral microbiota with CRC status for USA, MAL1, and MAL2 cohorts. Microbiome profiling with 16S rRNA gene sequencing was applied to tumor (CRC) and paired normal tissues (Normal) from CRC patients for all three cohorts, as well as healthy biopsies (Healthy Bx) for USA and MAL1

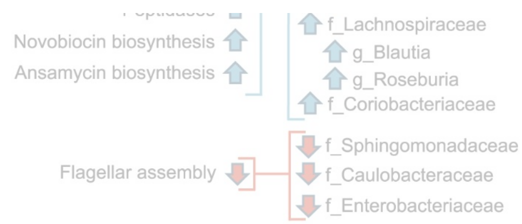






So...why do we care?

...And what did *I* do?



Goals:

1) Build original analyses of the MAL2 Cohort

2) Try to recreate analyses built by the authors

02

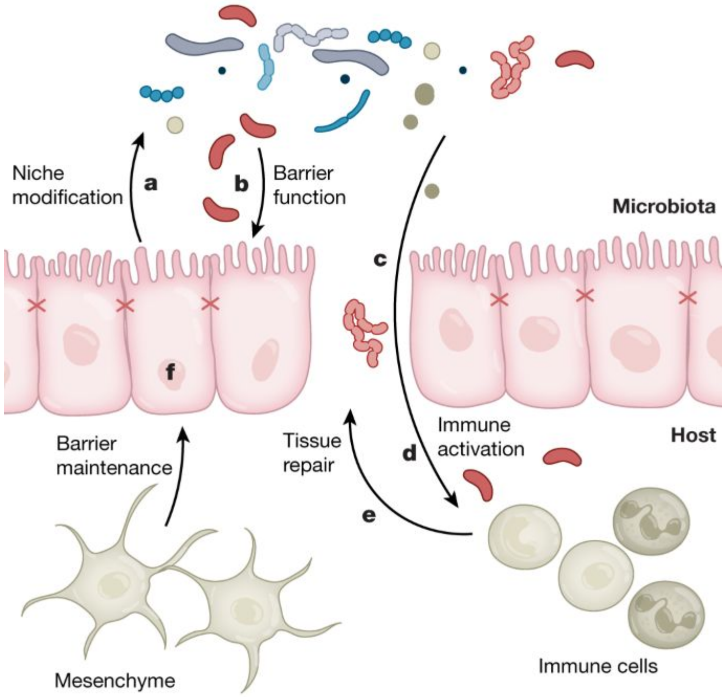
Ezra's 16S Amplicon Pipeline

Hypothesis Formation

Scientific question: Does gut microbial community composition and diversity differ between patients with colorectal cancer (CRC) and healthy individuals with normal colon tissue, and if so, which specific taxa drive these differences?

Null hypothesis (H_0): Gut microbial alpha diversity, beta diversity, and taxonomic composition do not differ between CRC and Normal colon tissue samples — any observed differences are attributable to inter-individual variation rather than disease status

Alternative hypothesis (H_1): Gut microbial diversity is reduced and community composition is significantly altered in CRC samples relative to Normal samples, characterized by the enrichment of oncogenic pathobionts (e.g., Fusobacterium nucleatum, Peptostreptococcus stomatis) and the depletion of protective commensals (e.g., Faecalibacterium prausnitzii), consistent with disease-associated dysbiosis in the tumor microenvironment



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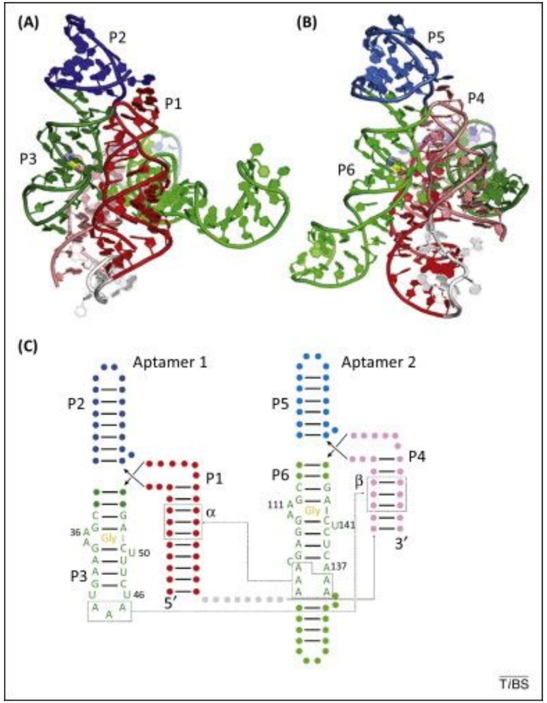
Dataset & Sequencing Strategy

Source: MAL2 cohort (Drewes et al., 2017; BioProject PRJNA325650) — surgical biopsy tissue samples collected from CRC tumor tissue and paired normal tissue taken from a non-cancerous region of the resected colorectal sample as far as possible from the tumor — 23 CRC patients yielding paired tumor + normal biopsies; 41 samples retained after QC in our analysis (20 CRC, 21 Normal)

Sequencing platform: Illumina MiSeq, paired-end 2×250 bp; targeting the V3-V4 hypervariable region of the 16S rRNA gene using primers 515F (GTGYCAGCMGCCGCGGTAA) and 806R (GGACTACNVGGGTWTCTAAT)

Bioinformatic pipeline: Primers removed with Cutadapt; quality filtering, denoising, and ASV inference with DADA2; taxonomy assigned against the SILVA v138 reference database; downstream analyses in R using phyloseq, vegan, and iNEXT



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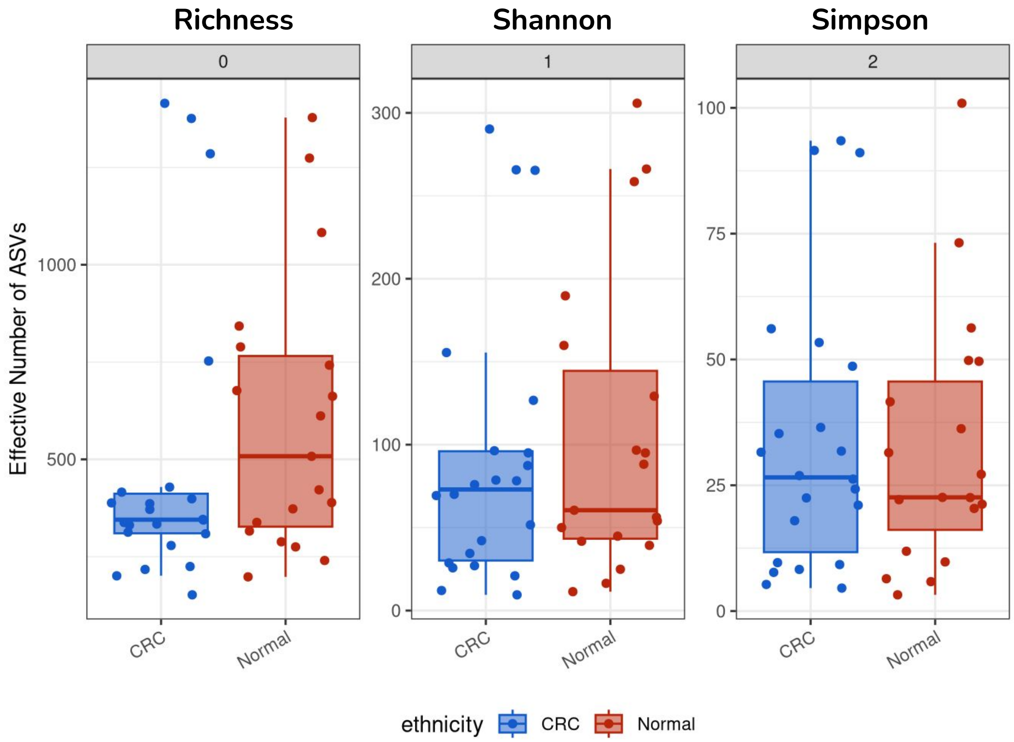
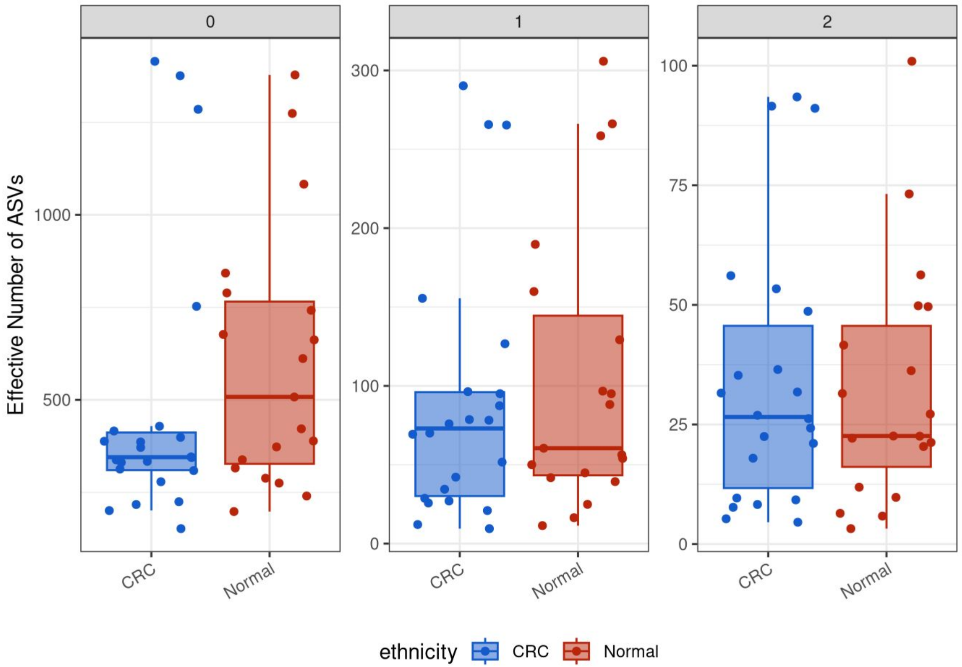
“Abundance Unweighted Counts is Significantly Lower in CRC Samples”



“Abundance Unweighted Counts is Significantly Lower in CRC Samples”



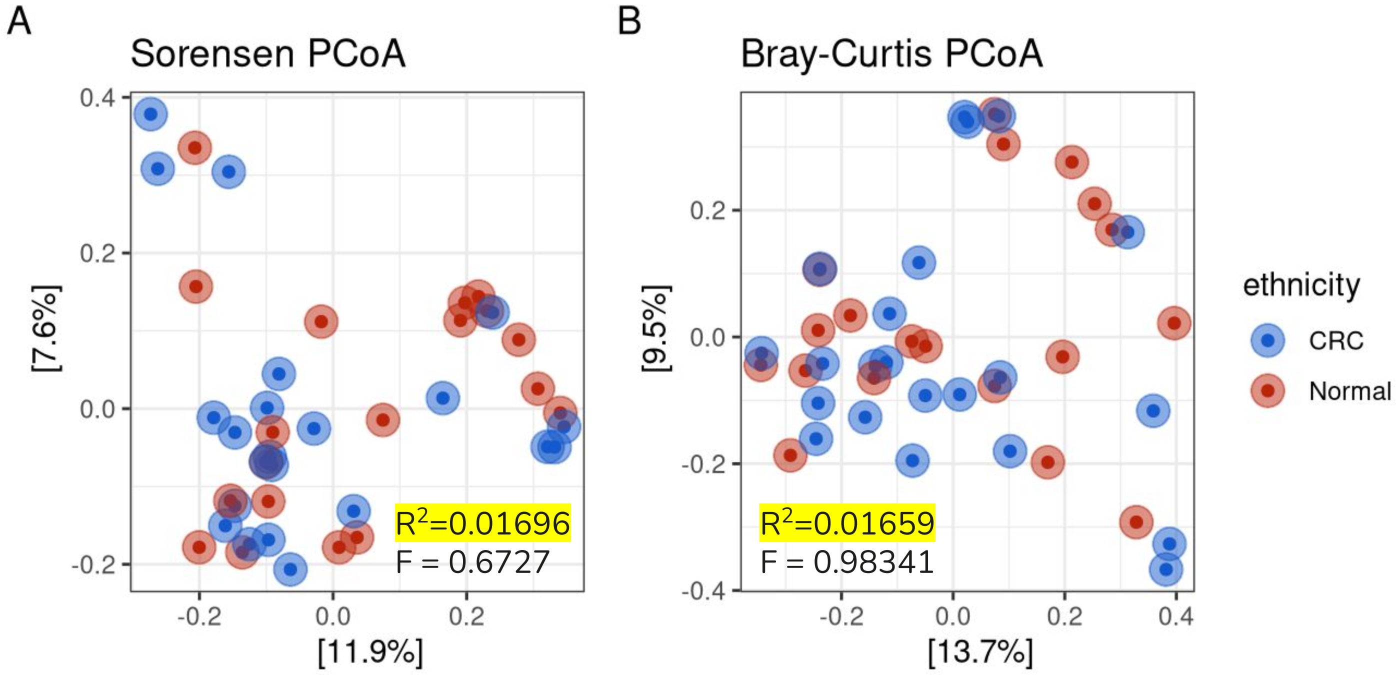
q = 0 (Richness): Normal samples show substantially higher richness than CRC, *consistent with the loss of rare commensal taxa during gut dysbiosis.*

q = 2 (Simpson): Dominant taxa are similarly structured between groups, suggesting CRC selects against rare taxa while preserving — or even enriching — overdominant ones.

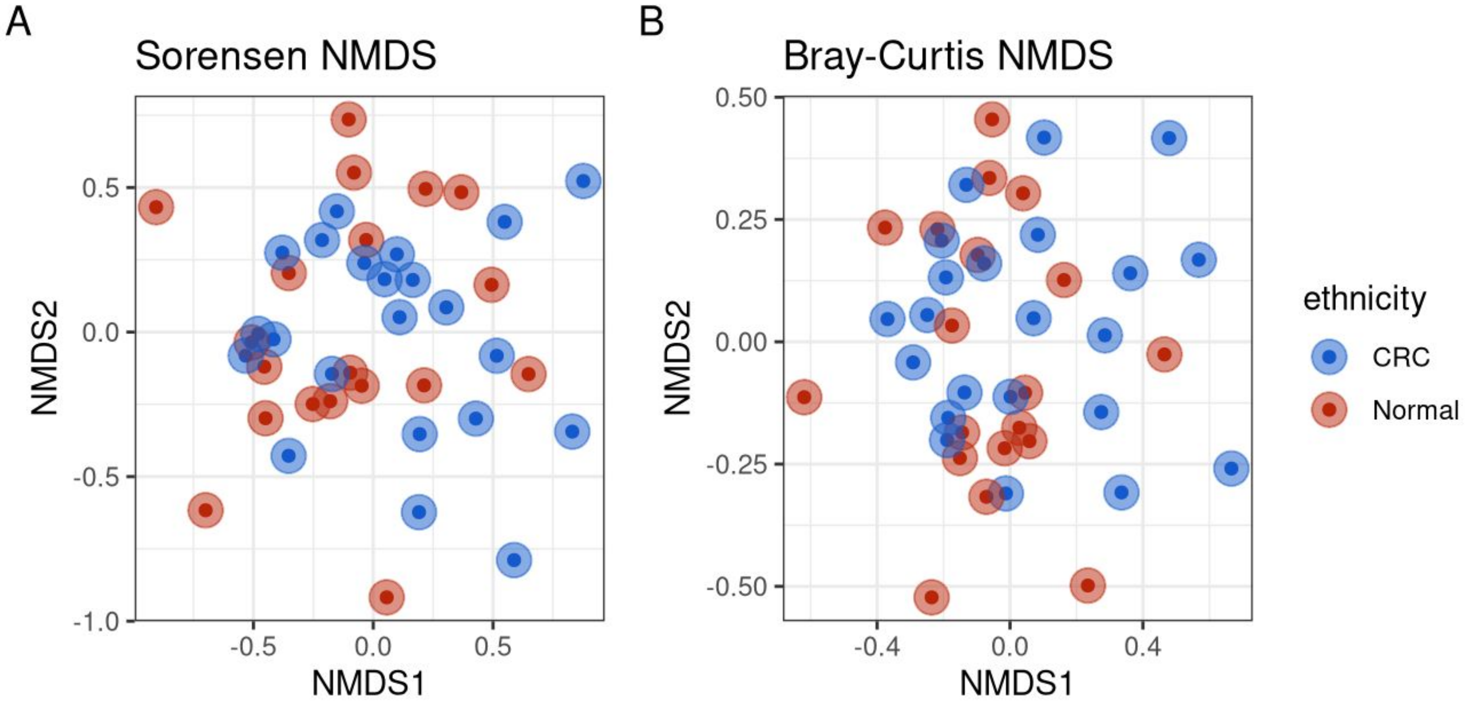
Biological relevance: This pattern *challenges* the idea that immunoprivileged tumor microenvironments allow for increased microbial density and the rise of opportunistic taxa that would otherwise be kept in check by host immunity.

Do we expect *community-wide* shifts in the composition of our microbiome samples?

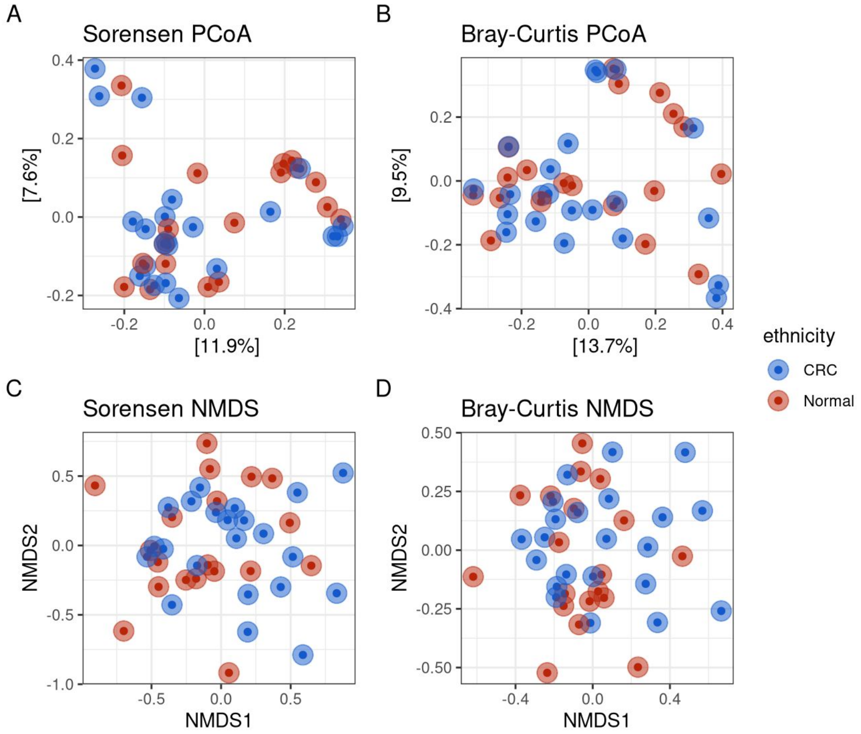
PCoA Analysis: Sorensen vs. Bray-Curtis



NMDS Analysis: Sorensen vs. Bray-Curtis



Why is this the case?



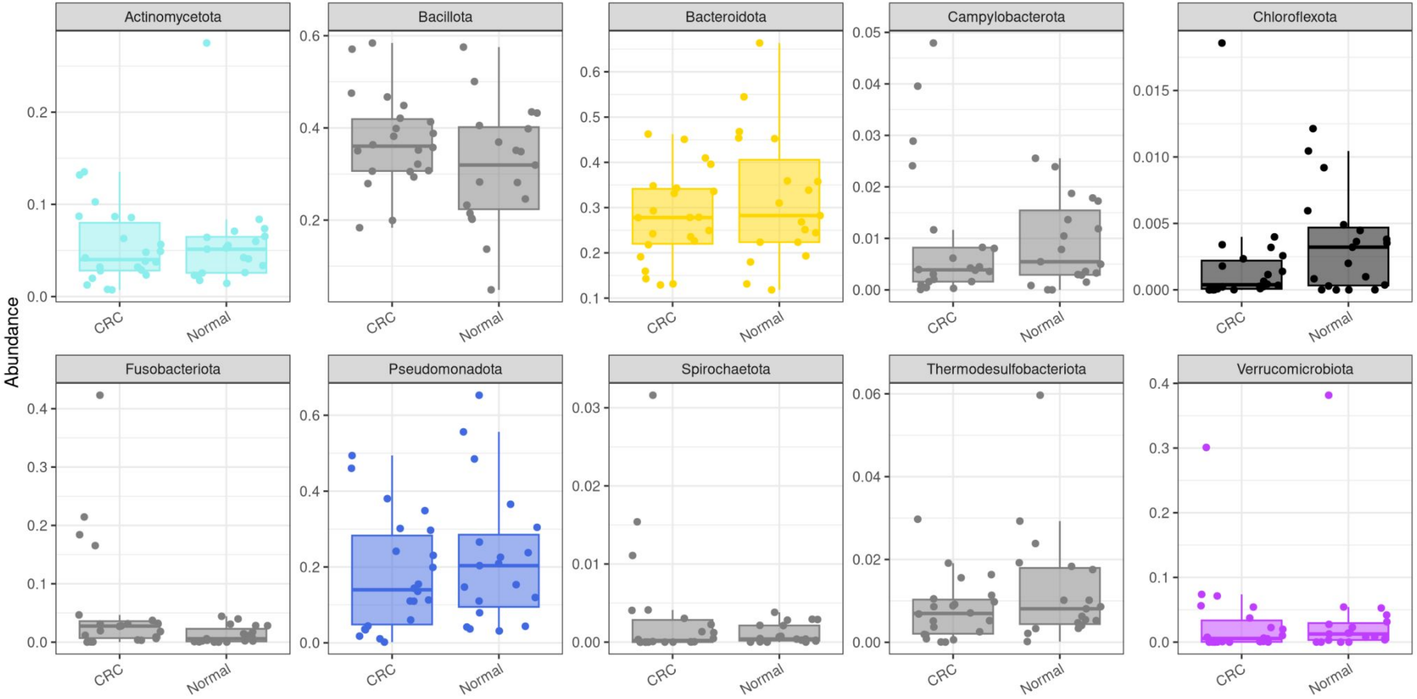
No clustering by CRC status: CRC and Normal samples are broadly intermixed across all four ordinations, indicating that CRC does not drive community-wide shifts in gut microbial composition.

Consistent across methods: Both PCoA and NMDS, using both Sorensen and Bray-Curtis, tell the same story, strengthening confidence in this conclusion.

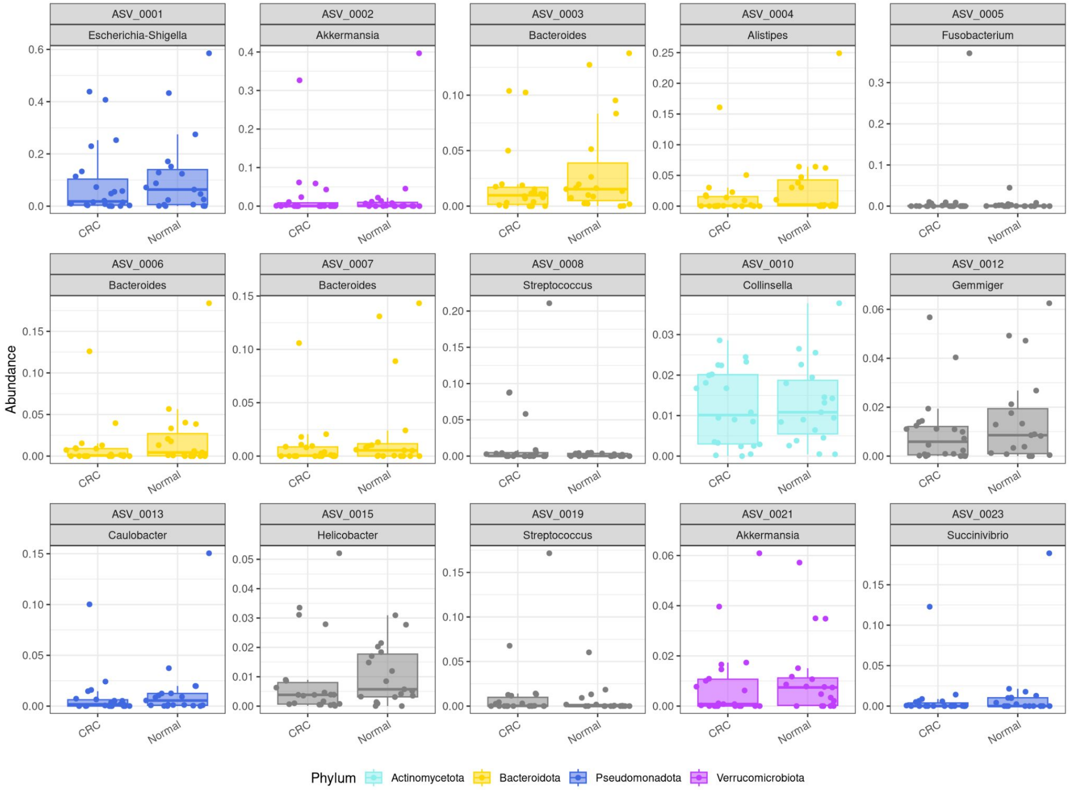
Inter-individual heterogeneity dominates: Host-specific factors appear to be the primary driver of community variation, consistent with the well-known high variability of the human gut microbiome across individuals.

Do we expect *heterogenous* shifts within individual microbial taxa?

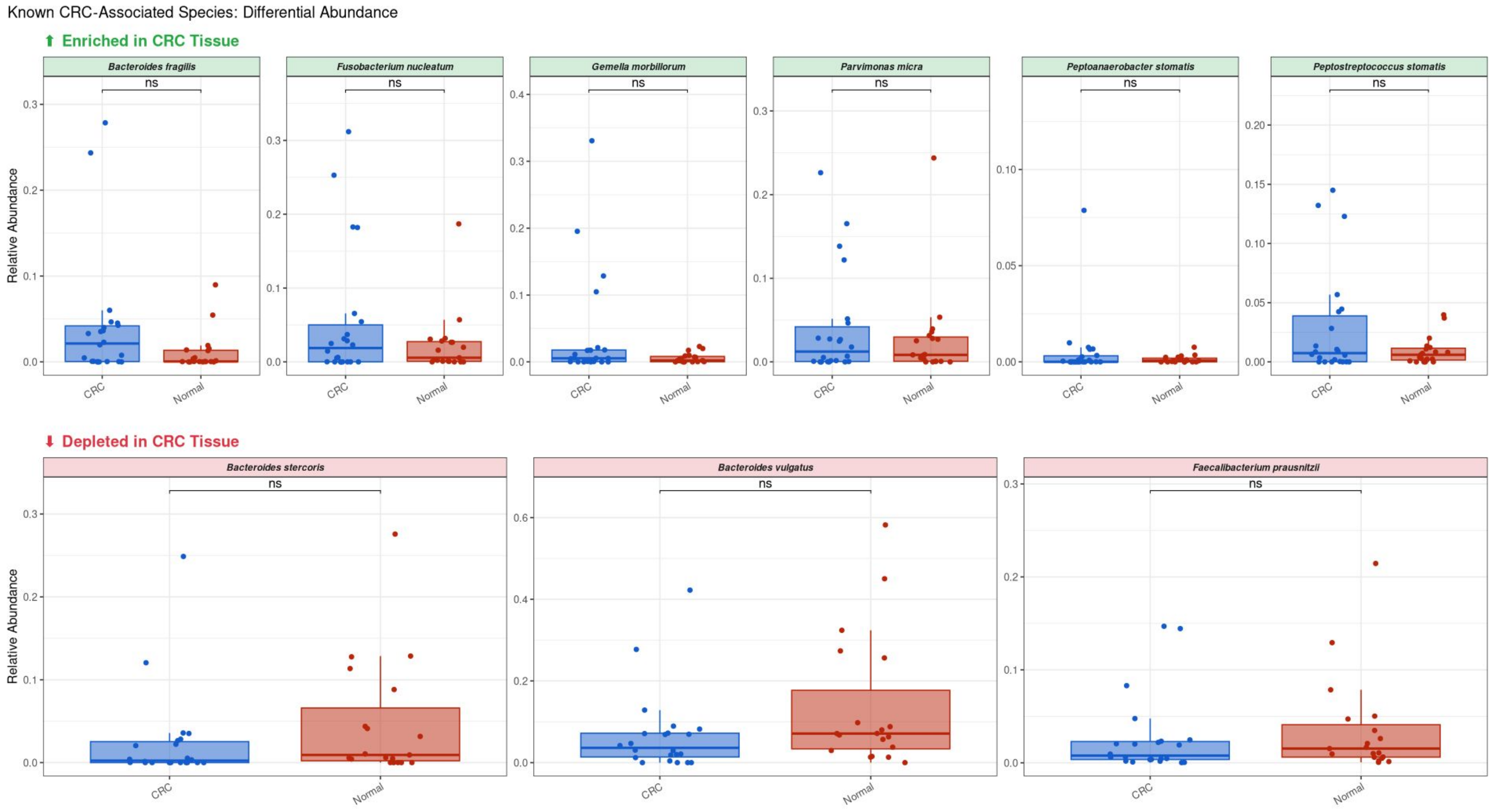
How about 10 most common *Phylum*?



How about 15 most common *ASVs*?



How about *canonical* species?



Future Directions...

Table S1 Characteristics of USA, MAL1 and MAL2 cohorts

Cohort	PatientID	Status	16S-Seq SampleID	Biofilm	Age	Gender	Race	Side	Tstage	Nstage	Mstage	Clinical Stage
MAL2	S034	Normal	S034NL	No	57	Female	Chinese	L	N	N	N	N
MAL2	S034	CRC	S034TL	Yes	57	Female	Chinese	L	T3	N0	M0	II
MAL2	S035	Normal	S035NL	No	74	Male	Indian	L	N	N	N	N
MAL2	S035	CRC	S035TL	No	74	Male	Indian	L	T3	N1	NA	III
MAL2	S036	Normal	S036NL	No	49	Female	Chinese	L	N	N	N	N
MAL2	S036	CRC	S036TL	No	49	Female	Chinese	L	T3	N2	M1	IV
MAL2	S037	Normal	S037NL	Yes	57	Female	Chinese	L	N	N	N	N
MAL2	S037	CRC	S037TL	Yes	57	Female	Chinese	L	T3	N1	M0	III
MAL2	S038	Normal	S038NL	Yes	49	Male	Malay	L	N	N	N	N
MAL2	S038	CRC	S038TL	Yes	49	Male	Malay	L	T3	N1	M0	III
MAL2	S039	Normal	S039NR	No	75	Female	Chinese	R	N	N	N	N
MAL2	S039	CRC	S039TR	Yes	75	Female	Chinese	R	T3	N0	M1	IV
MAL2	S040	Normal	S040NL	Yes	84	Female	Chinese	L	N	N	N	N
MAL2	S040	CRC	S040TL	Yes	84	Female	Chinese	L	T2	N0	M0	I
MAL2	S041	Normal	S041NL	Yes	50	Male	Malay	L	N	N	N	N
MAL2	S041	CRC	S041TL	Yes	50	Male	Malay	L	T3	N1	M0	III
MAL2	S042	Normal	NA	Yes	58	Male	Chinese	R	N	N	N	N
MAL2	S042	CRC	S042TR	Yes	58	Male	Chinese	R	T3	N2	M0	III
MAL2	S043	Normal	S043NL	No	52	Female	Malay	L	N	N	N	N
MAL2	S043	CRC	S043TL	No	52	Female	Malay	L	T3	N0	M0	II
MAL2	S044	Normal	S044NR	No	41	Male	Indian	R	N	N	N	N
MAL2	S044	CRC	S044TR	Yes	41	Male	Indian	R	T3	N0	M1	IV
MAL2	S045	Normal	S045NR	Yes	50	Female	Chinese	R	N	N	N	N
MAL2	S045	CRC	S045TR	Yes	50	Female	Chinese	R	T1	N0	M0	I
MAL2	S046	Normal	S046NL	Yes	52	Female	Chinese	L	N	N	N	N
MAL2	S046	CRC	S046TL	Yes	52	Female	Chinese	L	T3	N0	M0	II

References

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